

An Adjustable Simulation Model of Walt Disney World's "The Haunted Mansion"

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IDS 6145 Simulation Techniques

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1. Abstract

Project Overview & Key Findings

Simulation tools are valuable resources that can be used to optimize and operationalize a wide range of complex systems. Waiting queues at many of today's popular theme park attractions are often extremely long, tedious, poorly designed, and bottlenecked. In this research, we propose a method that examines a specific theme park attraction with an intricate level of detail designed to visualize a wide assortment of variability and complex parameters. Utilizing the AnyLogic personal learning edition (PLE) software, we have created a high-fidelity, realistic simulated prototype of Walt Disney World's The Haunted Mansion.

Major Recommendations & Methodology Summary

Our primary recommendation in creating this model is that it be used and implemented by amusement parks and similar venues for visualizing an accurate representation of attraction capabilities and maximum capacity. Although large-scale parks likely already have systems in place to analyze their rides, our recommendation is that this system be employed by other parks, smaller in scale, but with similar attraction objectives. Examples include county fairs, local state-run amusement parks, and circus attractions. In addition, we recommend that all parks experiencing longer queue times implement a fast pass or lightning pass system to divide their queues; further detail on this is presented below. Finally, this model can easily be applied to a wide assortment of rides; although it is specific in nature, it is easily tailorable to a wide range of attractions. Our model is best described as a throughput and queue dynamics simulation, which accounts for an accurate arrival rate, variability of ride passengers, a breakdown of lightning pass versus non-lightning pass users, ride time, and maintenance delays.

2. Introduction

Problem Statement

Modern amusement parks are growing in popularity and attract millions of visitors per year; guests continue to flock to these attractions to enjoy the thrills, narrative, and engaging nature of the rides. With a steadily increasing number of visitors year to year, wait times for individual attractions have proportionally increased, leading to longer queue times for rides. Extended wait times for rides cause problems for both visitors, theme park staff, and senior management. Visitors experiencing long wait times are more likely to experience dissatisfaction; many have traveled far distances to visit, with the added expenses of tickets, travel, and hotel costs. Visitor dissatisfaction leads to a decreased likelihood that the guests will return in the future, negatively impacting the revenue of the parks. Decreased revenue experienced by the parks lends itself to negative consequences affecting park employees, including downsizing, layoffs, and optimization shortcuts. Our aim in the creation of this project is to assist in resolving the issue of long wait times by optimizing the flow of passengers throughout the duration of an individual attraction.

System Description and Boundaries

Our system is a high-fidelity simulation prototype of Walt Disney World's "The Haunted Mansion", complete with adjustable parameters that provide an accurate representation of the entire ride from line entry through ride exit. The system consists of a queue divided into standard and lightning pass lanes with an accurate delay for ride duration and maintenance delays, followed by a ride exit. This is a standalone system designed to focus on the dynamics and processes of a single attraction; it does not account for human behavior variability or variables that may arise outside the ride, such as personnel shortages, weather-related park closures, or the opening of new rides, which may be more popular.

Research Objectives & Questions

Our primary objectives in the conduct of our research include the optimization of theme park queues and the comparison of lightning pass and standard queues. In addition, we are interested in the adjustment of system variables such as maintenance issues, peak hours, and the percentage of guests with lightning passes.

Report Organization By Section

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- 7) Experimental Design & Analysis

8) Conclusions & Recommendations

Appendices

- A) AnyLogic Simulation Model
- B) Model Presentation PowerPoint
- C) Average Replication Excel Data
- D) Python Analysis Scripts

3. Related Works

Relevant Work & Existing Models

Simulation of theme park queues, wait times, and popularity has been explored by a variety of researchers. In Taiwan, Chu et al. (2013) explored three rides and compared them by establishing no queue system for a control set and a comprehensive queue system for the evaluated dataset. In addition, the researchers were able to model customer satisfaction and found that it increased with the addition of a queuing system, yet they still acknowledged that the complexities of human behavior can be difficult to model, as customer choices are never uniform.

Oftentimes, rides and attractions will employ expedited lines available to customers for an additional fee, commonly known as fast pass lanes. Sakamoto (2020) examined the impact of fast pass lanes in Tokyo DisneySea, using a mathematical simulation model over 60 days. Sakamoto increased the number of priority entry guests based on a monotonously increasing function rather than hard caps on the number of lightning passes in the park and found that wait times were reduced. Another examination of the quick pass system was conducted by Guo et al. (2006), which compared the amount of free time that quick pass users had against the wait times of non-quick pass users. The researchers created a MATLAB model and found that, in general, the free time that quick pass users accumulate over a day is significantly longer than that of those without quick passes, indicating that a quick pass system is a good value for customers wanting to maximize their ride times and free time not spent in a line.

A constraining factor for retrieving accurate attraction wait times and reliable datasets is often the mobile or advertised data provided by the theme parks. Particularly useful to our research was a study conducted by Nirenberg et al. (2018), which found that the advertised wait times in the My Disney Experience mobile application were all rounded up in 5-minute intervals, indicating a slightly longer queue length than. The researchers developed a comprehensive algorithm to simulate and account for this delay, indicating in their results that the 5-minute rounding is detrimental to the queues because it discourages people from getting on rides.

Gap Analysis

Although a variety of studies exist for examining lightning pass queues and the movement of guests between rides in a theme park, there is a shortage of studies and published models that focus on individual attractions. Our analysis of this gap is that this information is likely proprietary and restricted by the intellectual property policies of the theme parks. Therefore, our model serves as a useful tool and a blueprint for anyone with existing data on their own attractions to quickly modify, implement, and make use of our tool.

4. Methodology

Conceptual Model Development

The Haunted Mansion is a unique attraction to study because the queue wait times are generally lower than most other park rides due to its continuous omni-mover conveyor belt system, which allows guests to continuously board and offload the ride. However, the ride historically also has numerous unplanned stops, ranging from short-term accessibility stops to accommodate wheelchairs, or longer maintenance stops to resolve ride issues. Visitors who enter the ride will pass through a set of gates where separate lightning pass and standard lanes are identified and marked for guests. Once guests reach the mansion, they are ushered into one of two “stretch rooms,” which provide a two-minute opening show to introduce visitors to a ride. Once the show is complete, lightning passes and standard lanes verge into a final boarding line, which culminates at the boarding of a “doom buggy” where the ride begins, lasting for approximately 8 minutes. Our simulation model accounts for an accurate number of visitors to the ride, standard and lightning pass lanes with queue waiting delays, a delay for time spent on the ride, and an exit for leaving the ride.

Model Assumptions & Limitations

Our model assumes a standard 14-hour operating day, with a total pool of 40,320 daily ride visitors; we also assumed a lightning pass percentage of 50% and a total capacity of 130 Doom Buggies, which were all set at either 2 or 3 passengers, and loading time was approximated with a single delay process. This model is limited in several aspects; it is not affected by individual guest behavior that may attract more visitors on certain days or times of the year, and it does not account for other situations such as ride evacuation scenarios or severe maintenance shortages.

Software Platform & Implementation Approach

This model was created and implemented inside AnyLogic, making use of the process modeling library for queues, sources, and delays, events for time-driven and conditional logic, and variables and parameters for tracking the state of the system. Our model can be described as a hybrid system, incorporating discrete event blocks for flow and queue behavior while using

system dynamics for a time-varying arrival rate, with random distributions for failure and repair times.

Data Collection & Pre-Processing Methods

Our dataset for the arrival rate of visitors to the Haunted Mansion was provided by an online resource, *Wartezeiten.app* (Wolf 2025), which hosts a collection of live waiting times over the course of months. The theoretical operational ride capacity (THRC) and operational ride capacity (OHRC) were sourced from a consolidated list of rides, estimated by an independent researcher Marco Wolfe, who has been counting and analyzing Disney World’s wait times for over 10 years. The mean time before failure for repairs was estimated based on around 10 total visits to the Haunted Mansion in 2025, where an average of 40 seconds per stop was noted, generally for wheelchair loading/unloading, but with a few longer stops up to 5 minutes where maintenance was required.

5. Model Development

Detailed Model Architecture

Our high-fidelity model begins with a source, comprised of a total source pool of approximately 40,320 daily ride visitors based on a standard 14-hour operating day at Magic Kingdom. This number is calculated by multiplying the Haunted Mansion’s hourly capacity by the number of hours the park is open for the day. The theoretical ride capacity (THRC) of the ride ranges from 2,400 to a maximum of 3,200 guests per hour. The Haunted Mansion utilizes a continuous loading system known as an omni mover, which allows for a constant stream of “Doom Buggy” ride vehicles to progress while visitors load the ride via this conveyor belt. In our scenario, a maximum capacity of 130 Doom Buggies is available to service guests.

A select output condition initially divides the source of visitors between lightning pass and non-lightning pass users, starting with a ratio of 50-50, but this can be fine-tuned and changed for further experimentation. Both lanes are then merged into the *rideGate* object, which represents the end of the queue, which occurs inside the stretch room preshow. Once visitors are inside, a ride delay begins, which accounts for buggy loading and the ride duration. Finally, a sink is employed to allow visitors to exit the ride, freeing up more Doom Buggies for additional guests; the experiment ends once it is cancelled or a threshold of 50,000 guests has been reached.

Entity Definitions & Attributes

RideVisitors (Source): This is the total pool of possible riders who are inside the park already and can potentially join the queue for the ride.

PassSplitter (Select Output): A conditional check that separates lightning pass from standard riders by an adjustable percentage.

StandardPass (Queue): A queue for those with standard access; these visitors did not pay extra for access beyond a park ticket

LightningPass (Queue): A queue for quick access, which enables guests' priority over standard access; these visitors paid a tiered fee for expedited ride entry.

StandardPassGate (Hold): This represents the holding capacity of the queue for standard access.

LightningPassGate (Hold): This represents the holding capacity of the queue for standard access.

rideGate (Hold): This merges the Lightning Pass and standard queues into the Haunted Mansion's doors, representing the end of the advertised wait time.

stretchroom (Delay): This is a standardized 2-minute delay where everyone goes into one of two rooms with a capacity of 50 persons each to watch a preshow.

loadingbuggy (Delay): This represents a delay where guests have completed the stretch room and are walking through the ride's walkway to the loading ramp for buggies.

Boarding (Batch): This represents loading the buggies in groups of two or three guests.

Haunted Mansion Ride (Delay): This is the actual ride time, represented by a distribution.

Deboarding (Unbatch): This represents getting out of the buggy and moving to the exit gates.

Ride Exit (Sink): Guests exit the ride.



Figure 1: AnyLogic model layout of the Haunted Mansion

Process Flows and Logic

1) Arrival and Routing

- Visitors arrive and are split into lightning pass and standard groups, joining the appropriate queue.



Figure 2: Haunted Mansion Stand-By & Lightning Entrance Adventure Is Out There. (2025).

2) Queue Batching

- Half of the riders are initially sorted into both standard and lightning pass queues.



Figure 3: Haunted Mansion Doors & End of Queue Adventure Is Out There. (2025).

3) Stretch Room Show

- Queues are combined at the doors of the Haunted Mansion, and all guests are delayed for 2 minutes during the stretch room preshow, which occurs in two separate rooms.



Figure 4: Stretch Room Preshow Adventure Is Out There. (2025).

4) Stretch Room Post Show Queue & Loading

- Guests merge into another line where buggies are loaded.



Figure 5: Stretch Room Post Show Queue Adventure Is Out There. (2025).

3) Ride Loading & Doom Buggy Dispatch

- Visitors are loaded onto doom buggies in either groups of 2 or 3, then the buggies are dispatched.



Figure 6: Stretch Room Post Show Queue Adventure Is Out There. (2025).

4) Ride Duration

- Once the buggies are released, the ride begins and proceeds until complete.



Figure 7: Haunted Mansion Ride Adventure Is Out There. (2025).

5) Ride Exit

- Simulation ends once 50,000 riders are processed, or the 14-hour day is complete.



Figure 8: Haunted Mansion Ride Exit Adventure Is Out There. (2025).

Parameter Estimation & Input Analysis

The dataset for our model from `Wartezeiten.app` contains 837 time-indexed observations of average guest waiting time with a mean of 48 arrivals per minute and a standard deviation of 9.18, indicating that the arrival rates do not fluctuate wildly. The shape of the observed dataset generally follows a Gamma fit distribution, with a slight skew to the right. Our dataset is not a good fit for an exponential distribution because the events occur randomly, and our histogram is not skewed right enough. A Poisson distribution is also not a good fit for our dataset because our data consists of continuous arrival rates per minute rather than integer event counts. In terms of p-values, our p-values are both less than 0.05 for exponential and Poisson distributions, further leading us to select Gamma.

We conducted an input analysis using a Kolmogorov–Smirnov (KS) goodness of fit test, and a Chi-Square goodness of fit test, and found the dataset to be an accurate fit for our experiment. In general, the dataset follows a predictable and stable pattern of arrival times rather than volatile arrival peaks, indicating that our arrival times are well validated for the model. As the queue time is observed to increase in the morning, stabilize in the middle of the day, and then decrease in the afternoon, a clear time dependence can be deduced in the arrival rate. Our model aims to address this through a time-dependent arrival rate, with a negative slope meaning that over time, λ (arrival intensity of riders) gradually decreases as the day goes on. This holds to be accurate because, in general, park visitors are much more likely to arrive as the gates open and the weather is cooler rather than later in the day. To model sudden shutdowns due to failures, it is possible to determine the periods during which failures have occurred and the length of time for the failures to be resolved using the historical data. Using Python, these periods were found, showing that the average time between failures occurring is approximately 23 hours, with an average repair time of 25 minutes. A probability distribution function of the observed failure

length and time between failures is created, showing behavior similar to an exponential distribution. This is confirmed with the inclusion of an exponential distribution using the average failure time and time between failures, showing that an exponential distribution aligns well with the observed data for modeling purposes.

===== INPUT ANALYSIS =====

Arrival samples count: 837

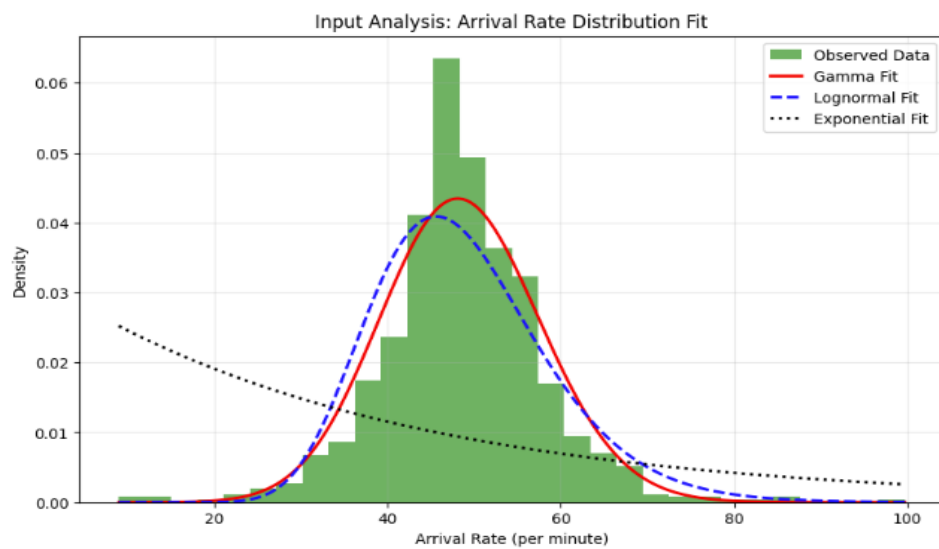
Arrival mean: 48.598, std: 9.181

--- Estimated Parameters (MLE) ---

Gamma: shape=390.268, loc=-132.978, scale=0.465

Lognormal: s=0.209, loc=0.000, scale=47.644

Exponential: loc=8.951, scale=39.647



--- KS Goodness-of-Fit ---

Gamma KS: stat=0.0564, p=0.009417

Lognormal KS: stat=0.0903, p=2.178e-06

Exponential KS: stat=0.4312, p=6.056e-142

--- Chi-square Goodness-of-Fit ---

Gamma Chi2: stat=266.3756, p=3.457e-52

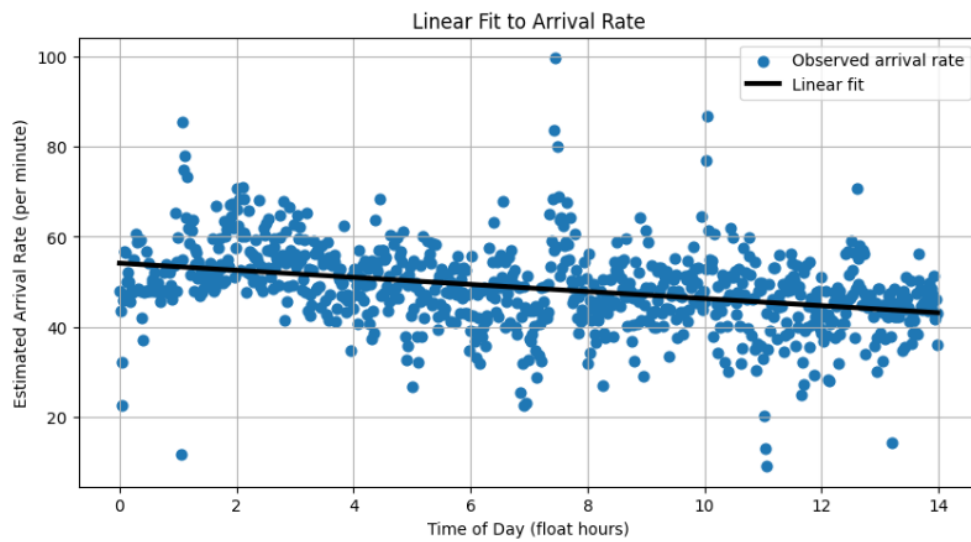
Lognormal Chi2: stat=11252.8031, p=0

Exponential Chi2: stat=1896.7745, p=0

FastPass Bernoulli Chi-square stat / p: 1.4400 0.2301

BuggyCapacity discrete-uniform Chi-square stat / p: 0.9216 0.3371

Figure 9: Input Analysis & Goodness of Fit Test Results



Slope: -0.7879026365678381
Intercept: 54.124770696667134

Figure 10: Linear Fit to Arrival Rate

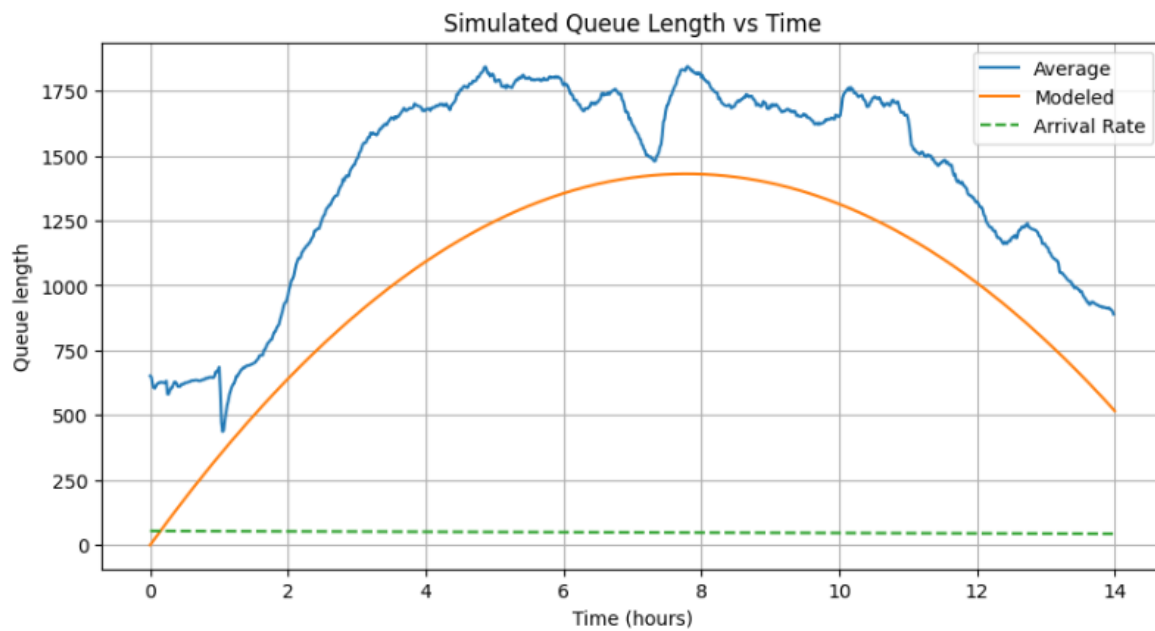


Figure 11: Simulated Queue Length Over Time

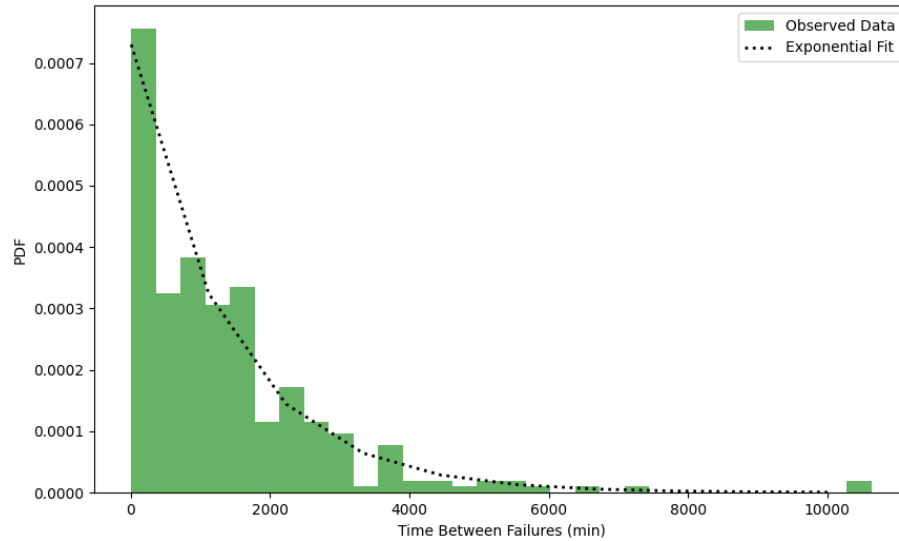


Figure 12: Observed and Modeled Distribution of Time Between Failures

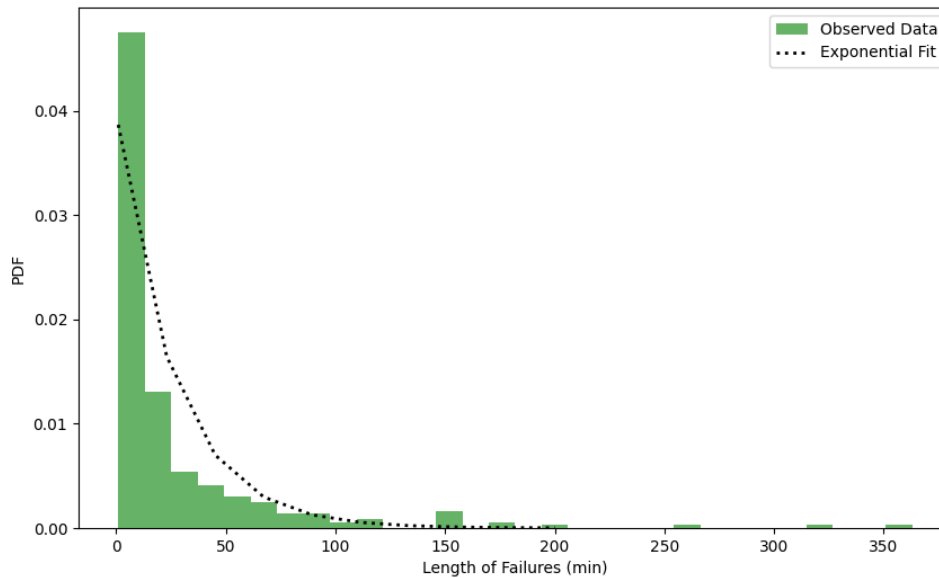


Figure 13: Observed and Modeled Distribution of Length Between Failures

6. Verification & Validation

Verification & Validation Techniques Used

Verification and validation of the model occurred with a statistically focused visit to the Haunted Mansion, where we found that our actual queue wait time, ride time, and number of breakdowns were consistent with our model's findings. In addition, our model's operational ride capacity was found to be 2,781 visitors per hour, which closely mirrored that of the OHRC found in Crookba's estimate of 2880 (Crookba's maximum capacity is 3200). In addition, according to the DarkRide Database, a reliable source for online theme park datasets, the Haunted Mansion's maximum throughput is 3,200 guests per hour. We attribute the difference between the maximum (3200) and our operational capacity (2,781) to wheelchair loading delays and

breakdowns. We have a high level of confidence in the model's accuracy based on our heuristics, visits to the ride, and independent online databases.

7. Experimental Design & Analysis

Simulation Experiments Conducted

In total, 200 Simulation runs were executed over the course of 20 replications to verify that the model is an accurate representation of the Haunted Mansion's queueing process. A series of key performance indicators (KPIs) was collected to observe a chosen set of observations to provide a variety of informative datapoints for park staff and senior management.

Sensitivity Analysis

To understand how the distribution of guests between the Normal line and the Fast Pass line affects overall ride performance, we ran a structured sensitivity analysis using AnyLogic's Parameter Variation Experiment. The key parameter varied was pNorm (Probability of Standard Pass holders), which represents the probability that a guest chooses the Normal queue (with 1-pNorm selecting the Fast Pass line). This parameter directly reflects guest behavior and crowd-mix scenarios that the ride might face throughout the day.

Experiment Setup

To initiate the experiment, we employed a varied parameter for the distribution of fast pass and standard ticket holders, using the nomenclature pNorm, with values of 0.1 to 1.0 with a step of 0.1. In plain text, this means 10-100% of visitors are using the standard pass, and between 0-90% of visitors are using the Lightning Lane for entry. Additional inputs to the model included parameters for throughput, slope, intercept, mean time before failure (MTBF), mean time to recover (MTTR), and batch size. For simulation replications, 20 iterations total per pNorm value were included to optimize randomness to capture an accurate representation of system behavior. Finally, the key performance indicators collected included the total guests served, queue lengths, ride cycles completed, delay patterns during breakdowns, and the utilization of both lane types.

Findings & Results Presentation

Key Observations

The sensitivity analysis focused on understanding how the model behaves when the proportion of guests choosing the Normal queue, represented by the parameter pNorm, is varied. We ran the model for ten different pNorm values ranging from 0.1 to 1.0. Each setting was simulated across twenty replications in order to capture the effects of randomness and to produce stable performance metrics.

Overall, the results show that changing pNorm affects the distribution of waiting guests across the queues but does not significantly influence the actual throughput of the ride. This outcome confirms that the true bottleneck is the ride capacity itself. Regardless of how guests are

divided between FastPass and Normal lines, the ride continues to process roughly the same number of people per hour.

As pNorm increases, a larger share of guests enter the Normal line. This naturally increases the length of the Normal queue while the FastPass queue becomes shorter. At pNorm values near one, the FastPass line is almost unused. The shapes of the queue curves remain similar across scenarios, suggesting that guest arrival behavior stays consistent and that pNorm only shifts how guests are allocated rather than changing the demand pattern.

Ride reliability metrics such as breakdown frequency and recovery time remain stable across all experiments because MTBF and MTTR parameters are unchanged. Higher pNorm settings lead to slightly more queue accumulation before a breakdown occurs, but the ride's operating cycle itself remains constant. The differences are visible in the queues rather than in the mechanical behavior of the ride.

The largest impact of pNorm is on the guest experience. When only a small percentage of guests choose the Normal line, FastPass users move quickly with very little delay, and the Normal line absorbs most of the waiting time. When pNorm is high, both lines operate similarly because FastPass demand is very low. This confirms that the FastPass system does not increase overall ride capacity. Instead, it redistributes waiting time across guest groups.

In summary, the sensitivity analysis shows that pNorm strongly influences how guests experience waiting but has little effect on throughput or ride downtime. This distinction is important because it highlights the difference between operational efficiency and service-level fairness. The ride continues to run at the same performance level, but the guest groups experience that performance very differently.

Performance Metric Analysis

The sensitivity analysis shows that the probability of guests choosing the Normal queue (pNorm) affects queue distribution but does not significantly change total throughput or ride cycle behavior. This confirms that the Haunted Mansion is a capacity-limited system: variations in guest behavior redistribute waiting time between Normal and Fast Pass lines but do not change how many riders are processed. Therefore, improving the queue mix does not increase throughput; it only reallocates where delays occur.

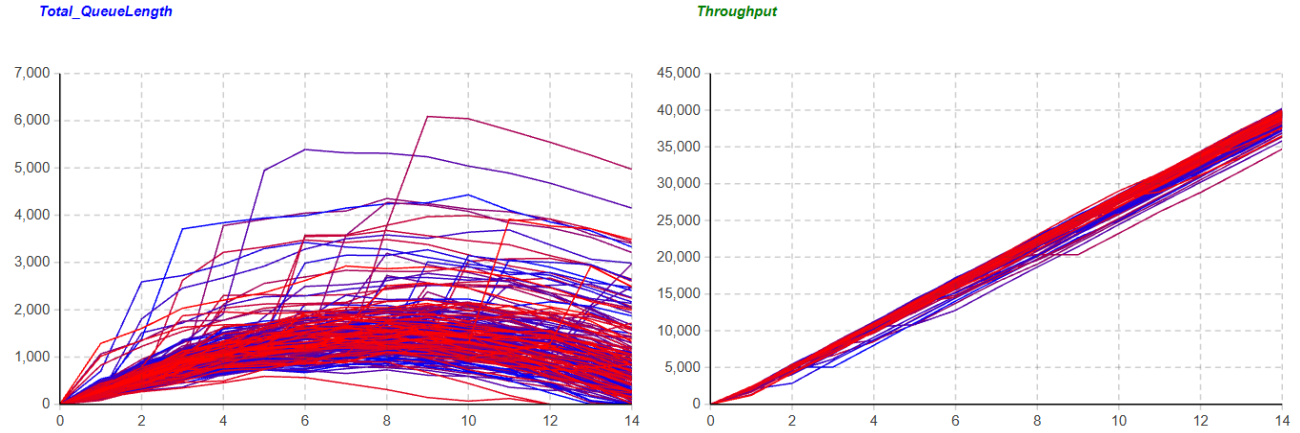


Figure 14: Total Queue Length Results & Throughput

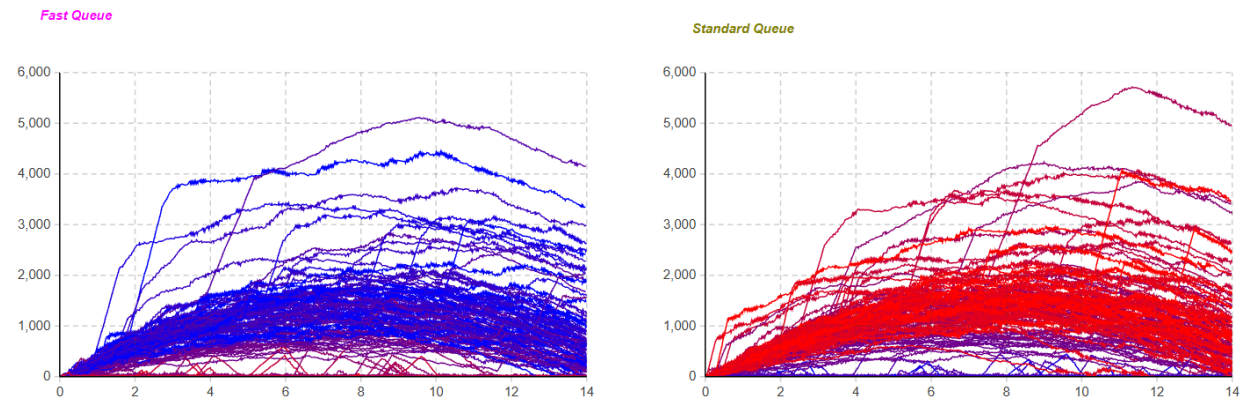


Figure 15: Fast Pass & Standard Queues

As was observed in the previous figures, there is a great deal of fluctuation that can be observed across our replications, as there will be variations in the arrival rate and the randomly occurring failures will halt all traffic passing through the ride. To mitigate these fluctuations in our analysis, an average was performed across all replications for each of the different parameter values for better visualization of the overall behavior. Due to difficulties in averaging the time series data of all of the replications for each parameter variation in AnyLogic, the data were imported into Excel, where the post-processing and plotting were performed. It is possible to observe the same behavior that was noticed previously, but the reduction in the amount of noise makes these behaviors far more prevalent. As the probability of a park visitor entering the normal queue increases, we see an overall increase in the normal queue length, a decrease in the Lightning Pass queue length, and no noticeable change in the throughput. Some variation can be observed in the averaged total queue length, but it does not appear to follow any trend based on the changes in the probability of a visitor entering the normal queue.

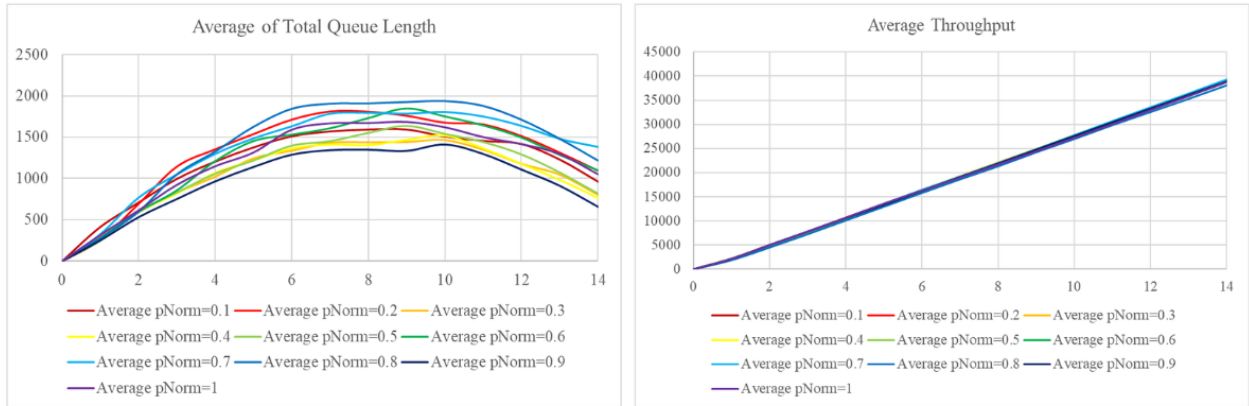


Figure 16: Averaged Total Queue Length Results & Throughput

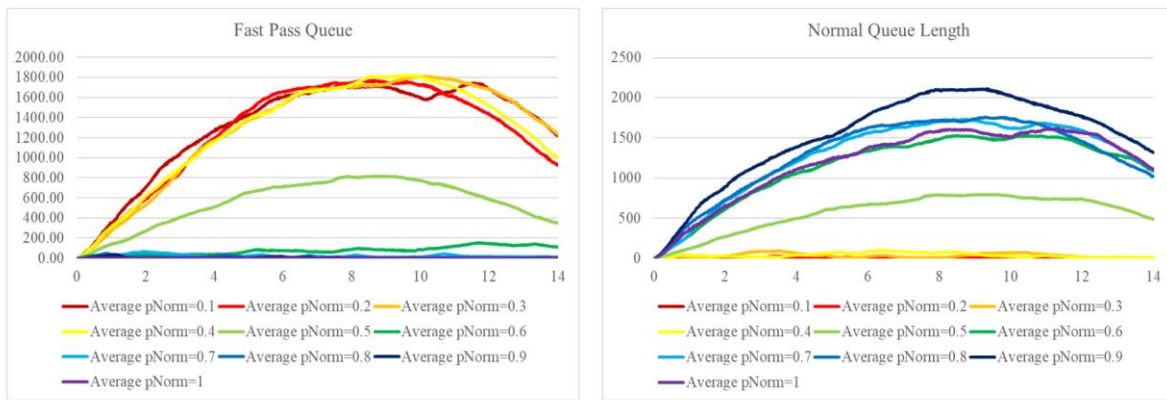


Figure 17: Averaged Fast Pass & Standard Queues

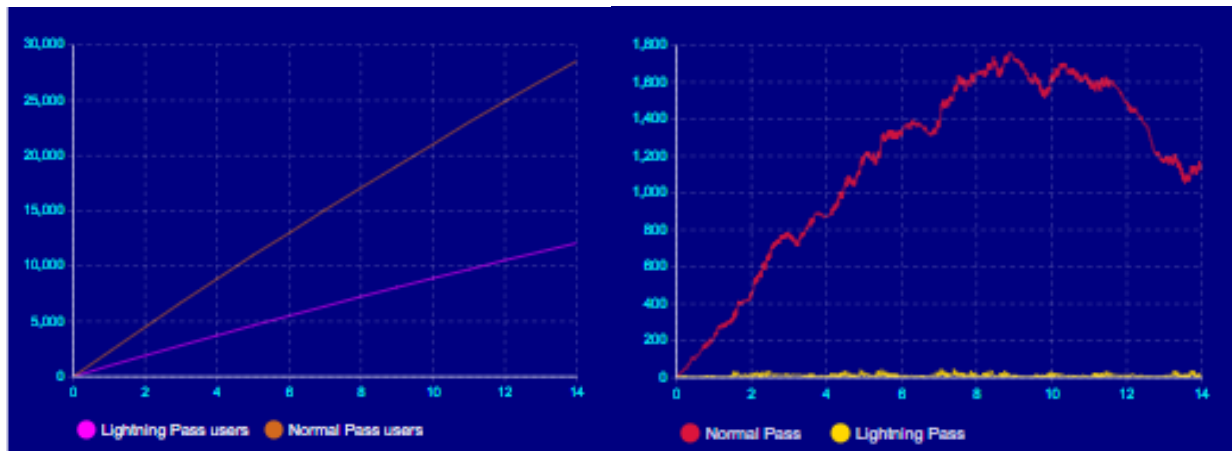


Figure 18: Sample Results After Simulation Run

8. Conclusions & Recommendations

Key Insights & Findings

Our model is a realistic portrayal of the Haunted Mansion's day-to-day operational activities; it easily meets demand later in the day but risks early overload issues in the morning. Any downtime experienced due to maintenance issues, which occur at a mean time before failure rate (MTBF) of 3 hours, causes spikes in the queue. These reliability issues are the largest factor in reaching our 2880 throughput target for each replication. In addition, the Lightning Lane system is for all intents and purposes, a redistribution method for wait times; it reallocates delays to those who pay, but overall, it does not eliminate the delays without a capacity increase for the attraction. This concept is further supported by the behavior of the standard lane, which absorbs almost all of the variability for arrival spikes, breakdowns, and recovery, and late afternoon spikes. In general, system performance for the Haunted Mansion is primarily driven by overall capacity, not the routing of guests between the Lightning Lane and standard.

Practical Implications

Our model can be practically applied to small-scale theme parks or local attractions curious about implementing a standardized queuing system, estimating throughput, or adding fast passes to their attraction. These metrics can drive critical business decisions, such as the requirement to add or remove employees to service the queue, costs to implement fast passes, and the costs and benefits of expanding the dimensions of a queue. We see two specific use cases for this model: a real-time dashboard for a specific ride and a tabletop tool for testing and experimenting with new processes and procedures. The framework of this model could be used in a live dashboard, where sensors along the queue and ride track the number of visitors in conjunction with maintenance delays and other closures. This live dashboard could be packaged in various ways to allow staff to check the flow and status of the ride quickly, rather than relying on radio communication or phone calls to employees in various regions of the ride. In addition, our model could also be used as a tabletop tool for experimenting with new procedures or projects. Examples of this could include changing the cost or availability of the lightning lane system, testing a new project to widen the queue size or improving the ride's loading system.

Recommendations for System Improvement

The system could be improved by implementing a more accurate measurement for the size of the queue in terms of distance. Often during peak hours, attraction workers will open overflow areas to accommodate high volumes of visitors; this logic is not present in our model. In addition, the interactions between other rides could also be modeled for a higher level of accuracy regarding available maintenance personnel; a serious issue at an adjacent ride could remove maintenance personnel from the ride and cause delays to repairs. In addition, our model

does not account for closing the queue for extended maintenance issues which cannot be immediately remedied so the severity of maintenance scenarios could be further explored and refined.

Limitations

Our research was limited in several areas, which prevented the project from reaching its true potential for implementation and real-world usage. Primarily, we were limited by a lack of real-world data for available maintenance personnel, total capacity of the ride, exact number of buggies, total park capacity, and other substantiating details, which are largely all proprietary and unavailable to the public. To counter this, most of our supporting data was crowd-sourced and estimated by frequent visitors to the ride. The arrival times are publicly available via the My Disney Experience mobile application, so our confidence in the accuracy of the wait times is high. Finally, student licensing in Any Logic PLE limited our research to 50,000 total agents present in the system. This cap prevented us from running the simulation over consecutive days as our total agents were consistently over 40,000 visitors per day.

Future Research

Future research in this area should examine the behavior of guests, adding agent-based modeling. Including these patterns of human behavior would add factors such as satisfaction, dissatisfaction, and likelihood to return, and the popularity levels of other attractions. These variables would contribute to an increased level of realism and accuracy for the model.

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